DATA VISUALIZATION PROJECT

Men’s T20 International Team & Match Performance

A cricket analytics dashboard built with HTML, CSS and JavaScript

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Project type	Data visualization and performance analysis
Data basis	Processed men’s T20I ball-by-ball records, as described in the dashboard

PROJECT SNAPSHOT

658 matches	**106** teams	**181,347** runs
8,561 wickets	**6,460** sixes	**148.1** avg. 1st innings

Prepared for academic project submission

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Overview

This project presents a web-based data visualization dashboard for men's T20 International cricket. It organizes match-level and ball-by-ball-derived summaries into headline indicators, year-by-year scoring trends, team profiles, a sortable team comparison table, dismissal counts, and venue scoring comparisons.

The dashboard states that it uses processed men's T20 international ball-by-ball CSV data. Matches without two completed innings are excluded. Its displayed headline summary covers 658 matches and 106 teams, with year labels spanning 2005–2026; the annual chart contains observations for the years present in the processed data.

Introduction

Cricket produces many performance measures: runs, wickets, run rates, wins, and venue-level scoring. A dashboard helps bring these measures together so a viewer can compare teams and spot patterns without reading each match record individually. This project focuses on descriptive analysis rather than prediction.

The dashboard is designed for readers who want a quick overview first, followed by deeper comparisons. Team selection updates the profile and phase charts, and the league table allows column sorting.

Objectives

Team performance

Compare match counts, wins, win percentage, average runs scored and average runs conceded.

Scoring trends

Review average first-innings scores and runs per over by year.

Innings phases

Compare batting run rate with bowling economy across overs 1–6, 7–15 and 16–20 for a selected team.

Dismissal patterns

Show the share/count of dismissal categories in the processed records.

Venue comparison

Compare average first-innings scores at grounds with six or more matches.

Interactive exploration

Let users select a team and sort the team table to explore the same data from different angles.

Dataset and dashboard overview

The dashboard's footer identifies its source as a processed men's T20 international ball-by-ball CSV. The source filename, original publisher, full column dictionary, and cleaning script are not included in the supplied HTML, so this report does not claim details beyond what the dashboard itself displays.

Displayed measure	Dashboard value
Matches	658
Teams	106
Runs scored	181,347
Wickets	8,561
Sixes	6,460
Fours	14,062
Average first-innings score	148.1
Batting-first wins	51.2%
Dashboard year range	2005–2026

Dashboard sections

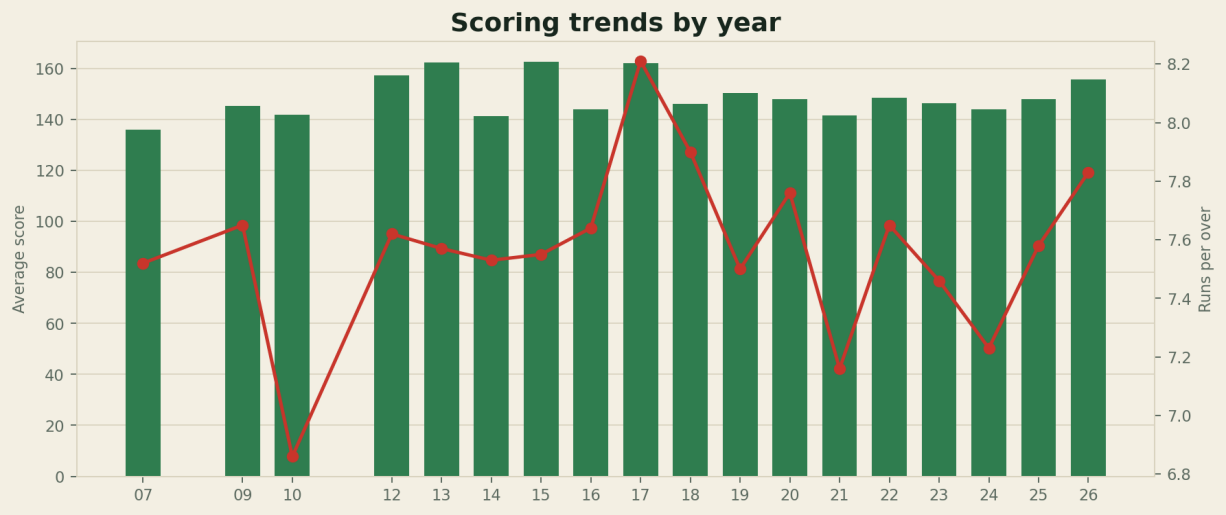
Headline KPI strip • scoring trend chart • selectable team profile • run rate by phase • win percentage by year • sortable team league table • dismissal breakdown • venue scoring comparison.

Analysis views and questions

The dashboard can be used to explore the following questions. These are analytical prompts supported by the displayed views, not claims that every question has a separate answer in the source data.

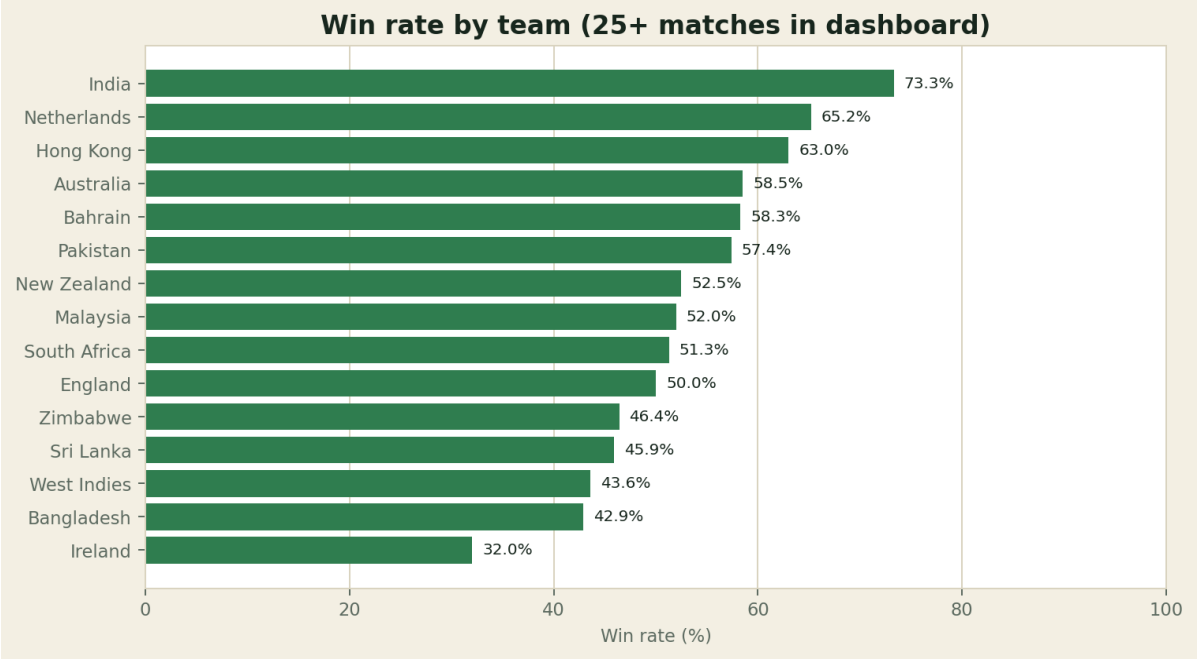
1. How does scoring change over time?

The annual chart uses bars for average first-innings score and a line for runs per over. The highest listed average first-innings score in the displayed annual series is 162.6 in 2015; the lowest listed is 136.0 in 2007. These values refer to the annual rows shown in the dashboard.



2. Which teams have the highest win percentages in the dashboard's team table?

The team table includes teams with 25 or more matches. In that displayed subset, India has a 73.3% win rate across 62 matches, followed by the Netherlands at 65.2% across 25 matches and Hong Kong at 63.0% across 27 matches. These are dashboard figures and should not be generalized to every team or every T20I match.



Approach / methodology

The supplied HTML indicates that the dashboard's summaries are already embedded as processed data in the page. The interface uses JavaScript to render the indicators and SVG charts in the browser. HTML provides the page structure, CSS controls the layout and visual theme, and JavaScript creates the chart content and interactive behavior.

1. Data scope

The dashboard describes its source as a processed men's T20I ball-by-ball CSV. It reports 658 matches and 106 teams. Matches without two completed innings are excluded. The dashboard notes that winners are determined from innings totals, so rain-affected results may differ from official records.

2. Summary measures

The headline strip presents totals for matches, teams, runs, sixes, wickets, average first-innings score and batting-first win percentage. Team profile values include matches, wins, win rate, average runs scored, average conceded, and win percentages when batting first or chasing.

3. Comparisons and interaction

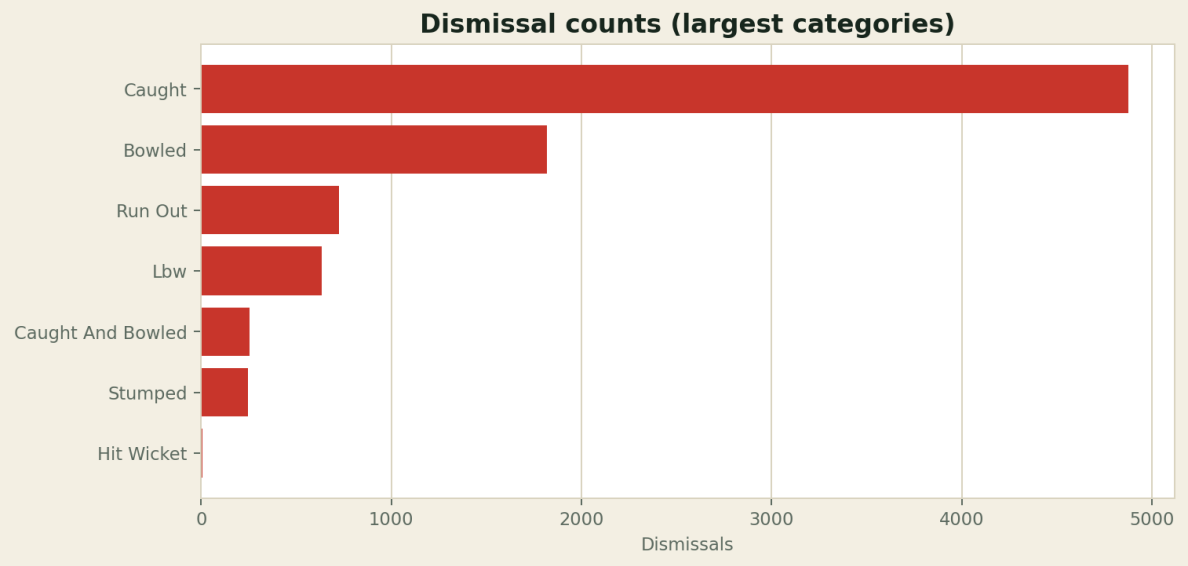
The team selector changes the profile values and charts. The league table can be sorted by team, matches, wins, win percentage, average score, average conceded, batting-first win percentage, and chase win percentage.

4. Chart design choices

Bars support comparisons between years, teams, dismissal categories and venues. A line is layered onto the annual scoring chart to show run-rate movement. Paired bars compare batting and bowling rates by innings phase. The dashboard uses a consistent cricket-inspired green, red, gold and blue palette.

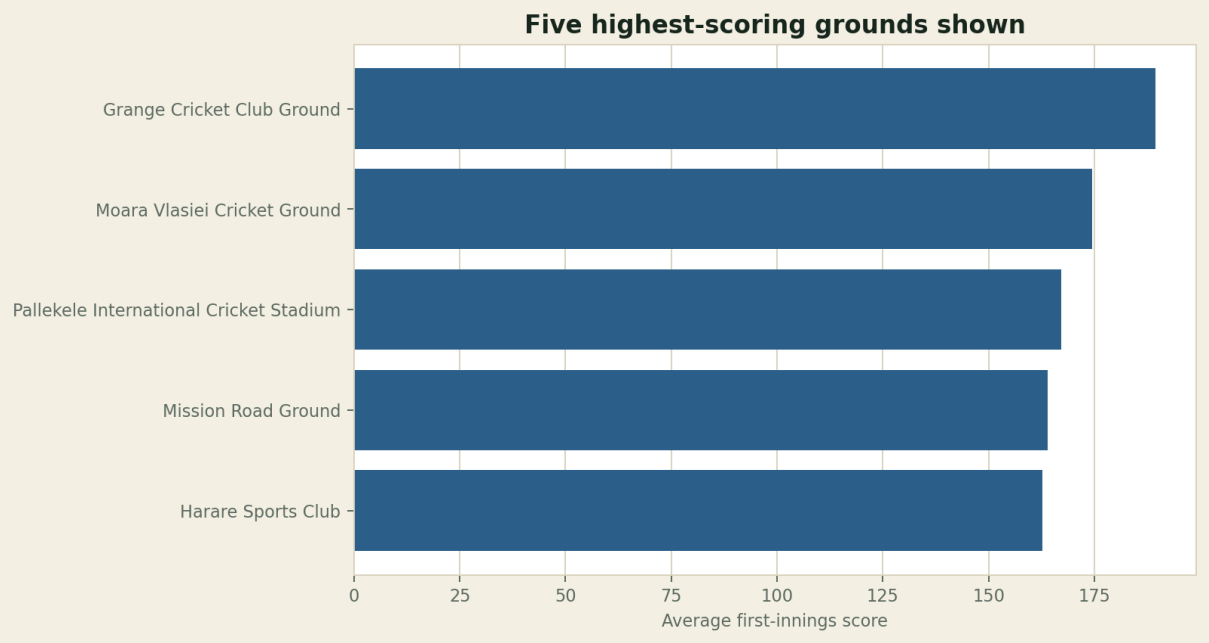
Dismissal patterns

Caught dismissals are the largest category shown: 4,874 out of 8,561 total listed dismissals. Bowled is next at 1,818, followed by run out at 725 and LBW at 633. The chart in the dashboard visualizes the largest dismissal categories.



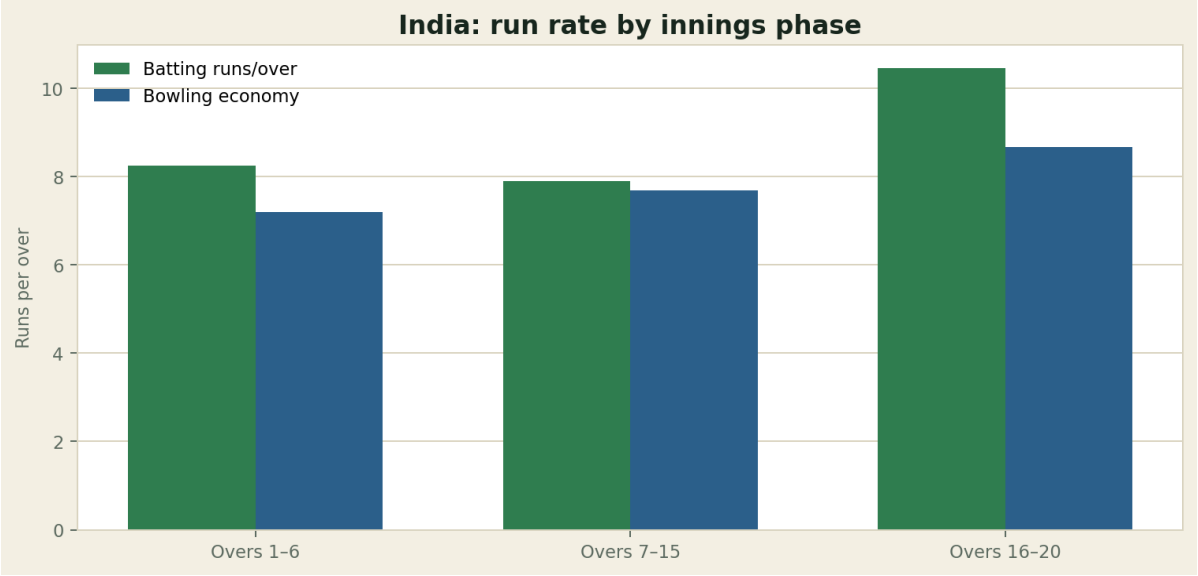
Venue comparison

The dashboard compares average first-innings score at grounds with at least six matches. Among the listed grounds, Grange Cricket Club Ground in Edinburgh has the highest displayed average at 189.5 across six matches. Sano International Cricket Ground is shown at 94.1 across seven matches. These venue averages should be interpreted with the match counts in mind.



Team phase profile: India

For India, the dashboard reports batting run rates of 8.24 in overs 1–6, 7.90 in overs 7–15, and 10.45 in overs 16–20. The corresponding bowling economy values shown are 7.19, 7.69 and 8.67. These are the phase metrics embedded in the dashboard's India profile.



Key findings

Headline scale

The dashboard summarizes 658 matches, 106 teams, 181,347 runs, 8,561 wickets, 6,460 sixes and 14,062 fours.

Batting-first results

The displayed overall batting-first win figure is 51.2%.

Team comparison

Among the teams shown in the 25+ match table, India has the highest displayed win percentage at 73.3%; the Netherlands is at 65.2% and Hong Kong at 63.0%.

Annual scoring

The highest listed annual average first-innings score is 162.6 in 2015; the lowest is 136.0 in 2007. The data chart includes only years present in the processed dataset.

Dismissals

Caught dismissals (4,874) are more numerous than bowled dismissals (1,818) in the dashboard summary.

Venue range

Listed venue averages range from 94.1 at Sano International Cricket Ground to 189.5 at Grange Cricket Club Ground, among the displayed grounds.

Skills demonstrated

- Data interpretation: reading team, match, scoring, dismissal and venue statistics.
- Web dashboard development: organizing a page with HTML and styling it with CSS.
- JavaScript visualization: rendering summary values, charts and interactive team selection/sorting.
- Visual communication: choosing chart forms that support comparison and trend reading.
- Analytical caution: interpreting percentages alongside match counts and noting exclusions and caveats.

Limitations and future scope

The supplied dashboard does not identify the original dataset URL, provide the raw CSV, document every cleaning rule, or include a validation comparison against official match results. As a result, the report describes only the values and methodology stated or implemented in the dashboard. The year range is shown as 2005–2026, while the annual chart contains rows for specific years where data is present.

Possible future improvements include adding a source citation and data dictionary, documenting data cleaning, providing filters for date range and match format, adding player-level batting/bowling views if the source data supports them, and validating rain-affected outcomes against official records.

Conclusion

This Men’s T20I Team & Match Performance project demonstrates how a web-based dashboard can turn processed cricket records into accessible summaries and comparisons. The dashboard brings together match KPIs, annual scoring trends, team win percentages, innings-phase run rates, dismissal categories and venue scoring averages in one interactive interface.

The displayed figures show a broad overview of 658 matches and 106 teams. Within the team table’s 25-or-more-match subset, India has the highest listed win rate, while caught dismissals are the most frequent listed dismissal category. Annual and venue comparisons also show that average first-innings scores vary across the rows and grounds displayed.

The project combines descriptive data analysis with front-end development using HTML, CSS and JavaScript. Its main value is to make the data easier to explore and communicate. All conclusions should be read with the dashboard’s exclusions and result-calculation caveat in mind.

References

1. User-provided dashboard: “Men’s T20I: Team & Match Performance” (HTML file).
2. Dataset description displayed in the dashboard footer: processed men’s T20 international ball-by-ball CSV; matches without two completed innings are excluded.
3. No external dataset URL or separate raw data file was provided with the dashboard.

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